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Road Segmentation

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ECS 174: Computer Vision
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Intro

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Goal: Automatically detect and segment roads from satellite images.

Motivation:

- Updating maps
- Faster disaster response
- Support autonomous systems

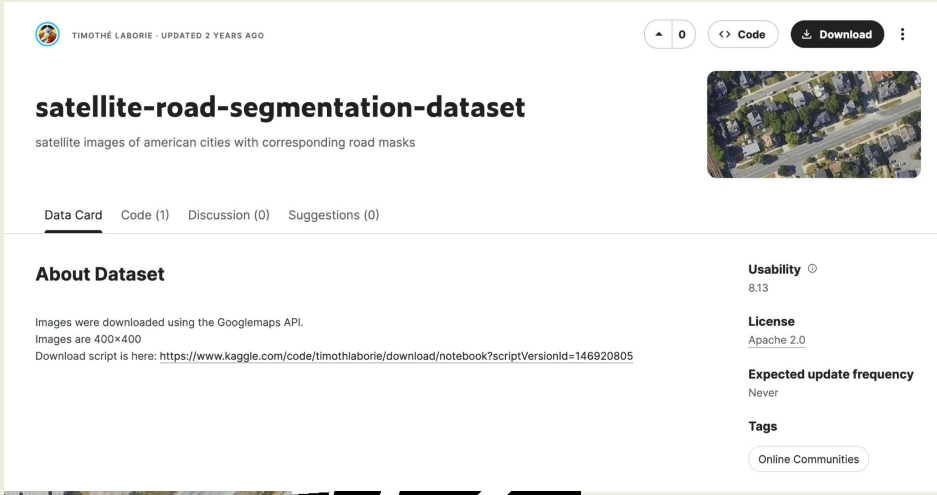
Dataset

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Source: Satellite Images for Road Segmentation (Kaggle)

Composed of 18297 400x400 px images of American roads in the cities of Los Angeles and Boston.

Each satellite image was paired with a ground truth image.





Data Preprocessing

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All images were split into 3 subsets:

- Training (70%)
- Validation (15%)
- Testing (15%)

Model Development

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- Standard U-Net encoder–decoder.
- Encoder uses 3×3 conv + ReLU + max-pool.
- Decoder uses 2×2 transposed convs and skip connections with encoder features.
- Skip connections from Encoder to Decoder.
- Final 1×1 conv outputs a single-channel road probability map.
- Model size: 31,031,745 parameters.

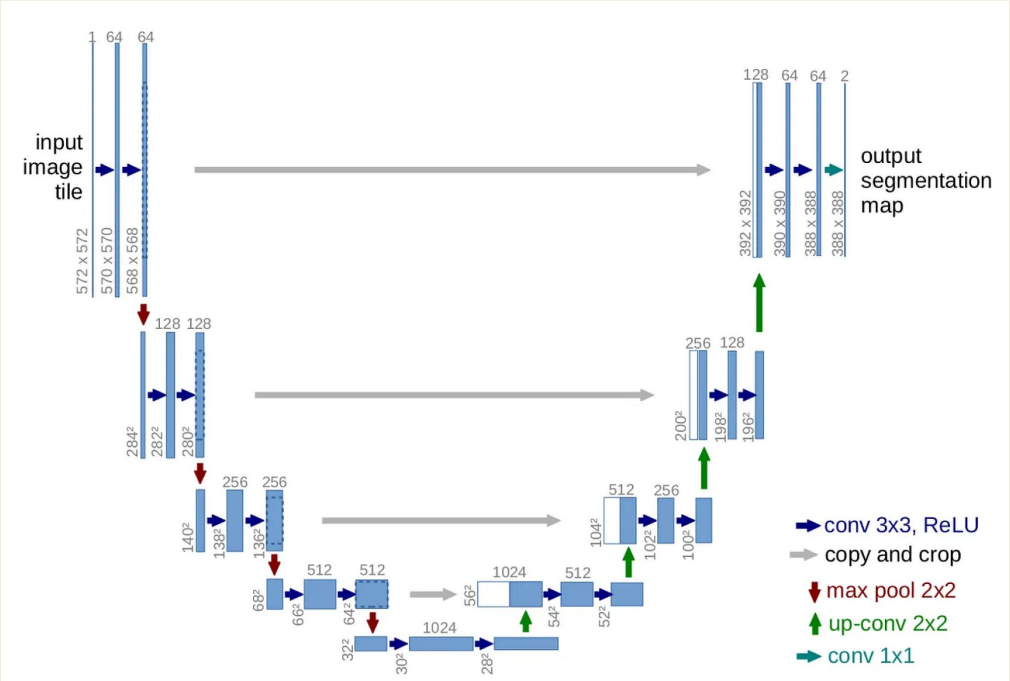


Image source: Itoaramendia, A. (2024). Medium.

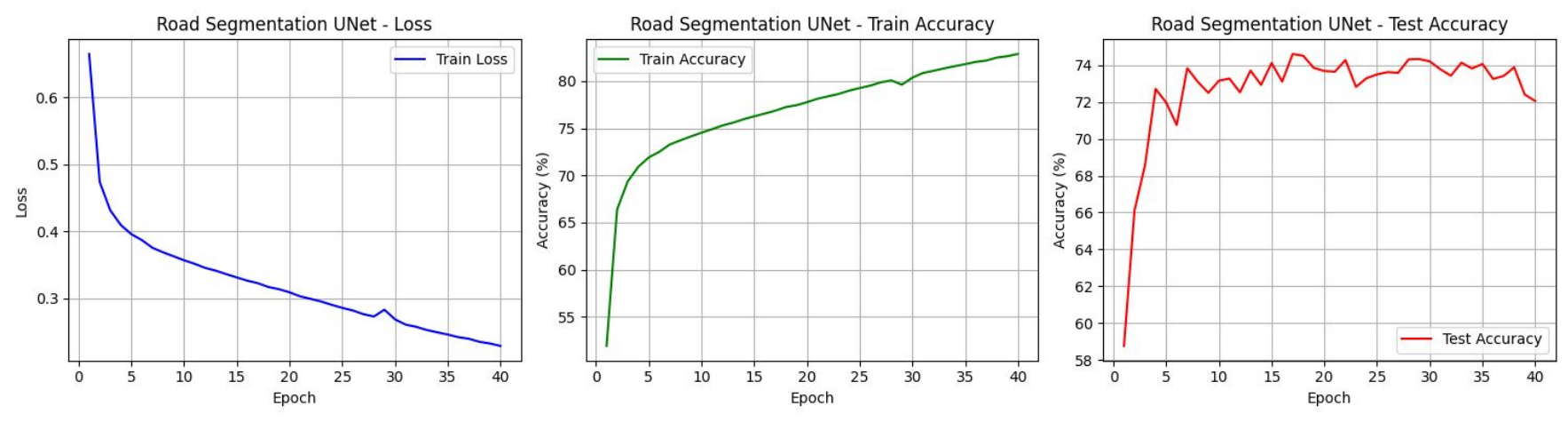
Training

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- Trained on CUDA, total time ≈ 4.5 hours.
- Adam optimizer, LR = 0.0005, batch size = 8, 40 epochs.
- Used BCE + Dice composite loss.
- Model outputs were sigmoid-activated and thresholded at 0.5.
- Tracked Dice coefficient for train/val performance; final Dice calculated on the test set.

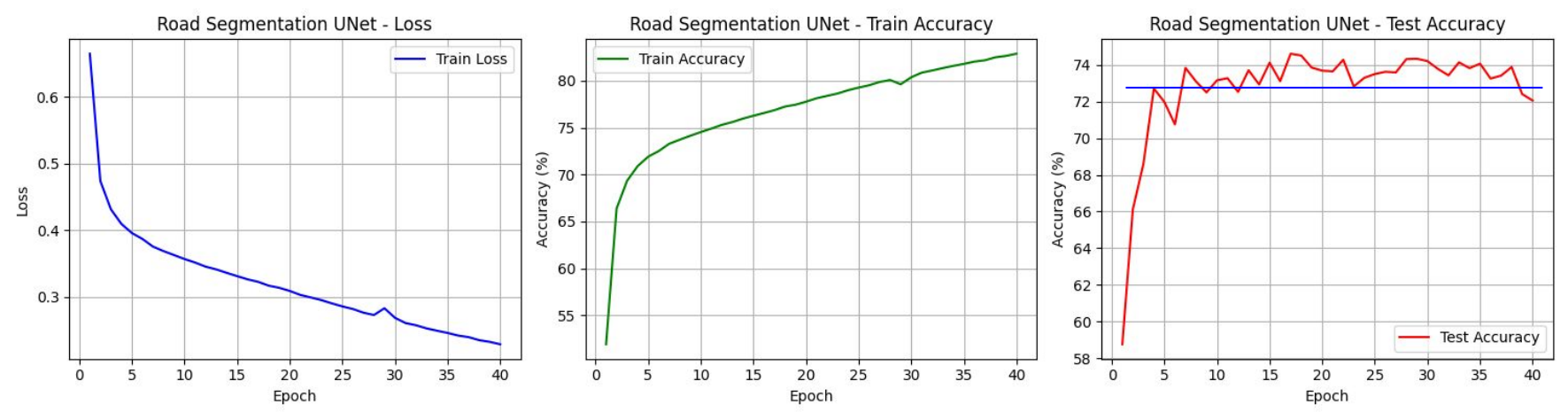
Training

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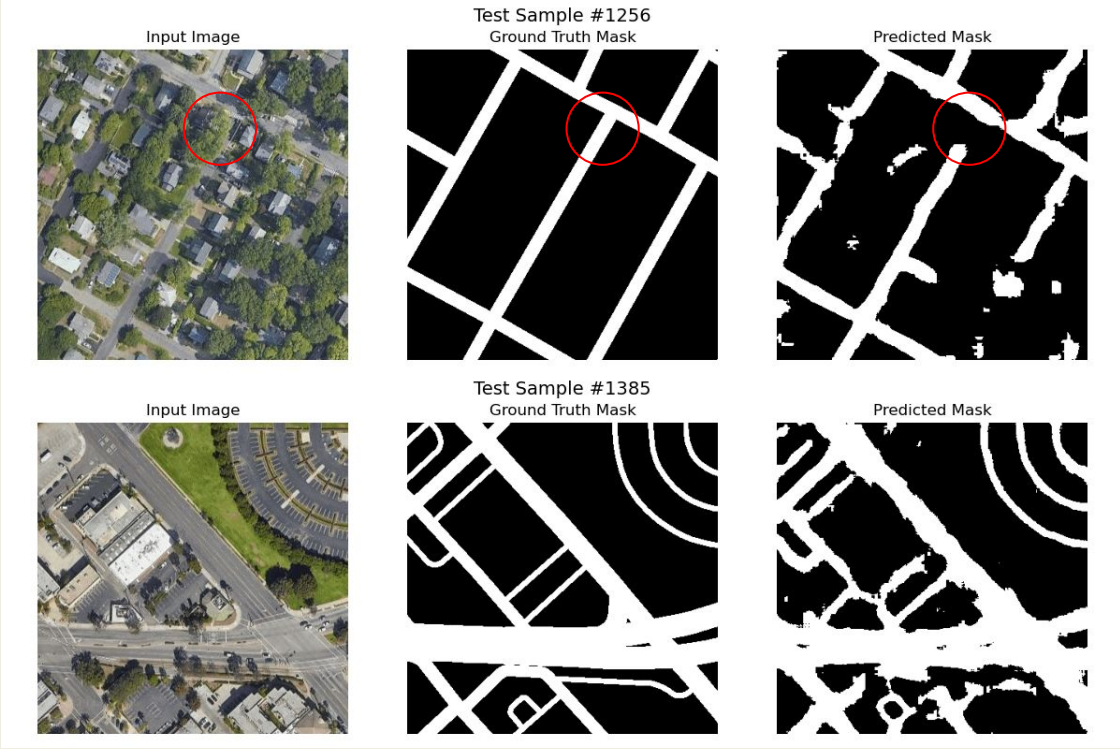
Results

- Steady improvement.
- Training loss decreased and training accuracy increased.
- Test accuracy plateaued at ~72–74%.



Results

- Checked predictions visually by comparing the model's masks to the ground truth.
- Final model test Dice coefficient of 72.26%



Discussion

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- Captures major roads well.
  - struggles with thin/ambiguous segments.
- Mild overfitting observed.
- Could improve with U-Net variants or more diverse data.
- Performance is sufficient for tasks needing broad road coverage.

Conclusion

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- U-Net successfully segmented roads from satellite imagery.
- Good foundation for future improvements.
- Shows that computer vision can support scalable mapping tasks.

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Thank You for Listening!